

EFK Stack on Minikube (Elasticsearch + Fluent Bit + Kibana)

This guide installs a complete logging stack on Kubernetes (Minikube):

- Elasticsearch (log storage + search)
 - Kibana (UI dashboard)
 - Fluent Bit (log collector)
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1. Prerequisites

Minimum system requirements

- CPU: 4+ cores (you have 12 ✓)
- RAM: 8GB+ (you have ~11GB ✓)
- Disk: 20GB+ free ✓

Tools required

- Minikube
- kubectl
- Helm
- Docker driver for Minikube

Check:

```
minikube version
kubectl version --client
helm version
```

2. Start Minikube

```
minikube start --cpus=4 --memory=8192
```

Verify:

```
kubectl get nodes
```

3. Create Namespace

```
kubectl create namespace logging
```

4. Install Elasticsearch

Add Helm repo

```
helm repo add elastic https://helm.elastic.co
helm repo update
```

Create values file

```
mkdir -p ~/efk-lab
cd ~/efk-lab

cat > elasticsearch-values.yaml <<EOF
replicas: 1

minimumMasterNodes: 1

resources:
  requests:
    cpu: "500m"
    memory: "1Gi"
  limits:
    cpu: "2"
    memory: "2Gi"

esJavaOpts: "-Xms1g -Xmx1g"

volumeClaimTemplate:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 5Gi
EOF
```

Install Elasticsearch

```
helm install elasticsearch elastic/elasticsearch \
  -n logging \
  -f elasticsearch-values.yaml \
  --wait --timeout 15m
```

Verify

```
kubectl get pods -n logging
kubectl get pvc -n logging

Test:

kubectl port-forward -n logging svc/elasticsearch-master 9200:9200

curl -k https://localhost:9200 -u elastic:<PASSWORD>
```

Get password:

```
kubect! get secret -n logging elasticsearch-master-credentials \
-o jsonpath="{.data.password}" | base64 -d; echo
```

5. Install Kibana

IMPORTANT (avoid Helm issues)

Always use:

```
helm install kibana elastic/kibana \
-n logging \
--wait --timeout 15m \
--set resources.requests.memory=512Mi \
--set resources.limits.memory=1Gi \
--set elasticsearchHosts="https://elasticsearch-master:9200"
```

Verify

```
kubect! get pods -n logging
```

Access Kibana

```
kubect! port-forward -n logging svc/kibana-kibana 5601:5601
```

Open:

<http://localhost:5601>

Login:

- user: elastic
 - password: (same as Elasticsearch secret)
-

6. Install Fluent Bit (Log Collector)

```
helm repo add fluent https://fluent.github.io/helm-charts
helm repo update
```

Install:

```
helm install fluent-bit fluent/fluent-bit -n logging
```

Verify:

```
kubect! get pods -n logging
```

□ 7. Generate Test Logs

```
kubectrl run log-test \  
  --image=busybox \  
  --restart=Never \  
  -- sh -c 'while true; do echo "EFK test log $(date)"; sleep 2; done'
```

8. View Logs in Kibana

1. Open Kibana:

<http://localhost:5601>

2. Go to:

Discover → Create Data View

3. Use index:

fluentbit-*

4. Set time range:

Last 15 minutes

5. View logs 

□ 9. Cleanup (optional)

```
helm uninstall fluent-bit -n logging  
helm uninstall kibana -n logging  
helm uninstall elasticsearch -n logging  
kubectrl delete namespace logging
```

Common Issues

1. Helm timeout

Fix:

```
--wait --timeout 15m
```

2. Kibana stuck

Delete job:

```
kubectrl delete jobs -n logging --all
```

3. Empty Kibana logs

Check Fluent Bit:

```
kubectrl logs -n logging <fluent-bit-pod>
```

Architecture

Kubernetes Pods



Fluent Bit (log collector)



Elasticsearch (storage)



Kibana (visualization)

Done

You now have a production-style logging pipeline on Kubernetes (Minikube).